

Bioinspired Clustering Algorithm for Credit Card User Segmentation Based on Mycelial Networks

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Abstract: Bioinspired optimization algorithms are increasingly applied in analyzing consumer behavior, particularly in financial datasets such as credit card transactions. This paper introduces the Fungi Kingdom Expansion (FKE) algorithm, a metaheuristic inspired by fungal growth, to optimize consumption patterns in large datasets. By simulating chaotic and deterministic fungal expansion, FKE effectively explores spending patterns, uncovering insights such as consumption trends and anomalies. In comparison to traditional methods like Particle Swarm Optimization (PSO), FKE excels in balancing local and global searches, making it highly suitable for optimizing credit card purchasing behaviors.

1 Introduction

In recent years, bioinspired algorithms have emerged as a crucial tool for solving complex optimization problems due to their effective balance between local and global search strategies [Lin et al., 2021]. The Fungi Kingdom Expansion (FKE) algorithm, inspired by the chaotic yet deterministic expansion of fungal mycelium, exemplifies this approach [Alnahwi et al., 2021]. Fungi expand their hyphae to find optimal conditions like moisture and temperature, and this adaptive strategy can be modeled computationally to efficiently explore search spaces [Fuchigami and Abreu, 2023].

The FKE algorithm employs a combination of chaotic (local) and deterministic (global) search mechanisms to optimize solutions iteratively [Alnahwi et al., 2021]. This approach mimics the immobile and mobile mass expansion observed in the fungal growth process [Zhao et al., 2022]. The immobile mass corresponds to the structure of the hyphae, while the mobile mass represents the flow of nutrients to the optimal tips of these hyphae [Huang et al., 2023]. Such a mechanism makes FKE particularly effective in avoiding local optima, a common problem in traditional optimization algorithms [Lin et al., 2021].

Fungi-inspired algorithms have demonstrated their potential in various engineering applications, such as antenna array optimization, where the FKE has shown superiority over traditional algorithms like Particle Swarm Optimization (PSO) and Genetic Algorithms (GA) [Alnahwi et al., 2021]. These bioinspired approaches, including Teaching-Learning-Based Optimization (TLBO), have been widely applied to solve large-scale industrial problems, including job shop and flow shop scheduling

[Fuchigami and Abreu, 2023].

The adaptability of fungi-based models has opened new avenues in optimization, making them applicable to both single-objective and multi-objective optimization problems [Zhao et al., 2022]. For instance, the chaotic and random dispersion modes used by the FKE have proven to be effective in addressing complex problems where traditional algorithms may struggle to maintain a balance between exploration and exploitation [Lin et al., 2021].

2 State of the Art

The Fungi Kingdom Expansion (FKE) algorithm represents a cutting-edge bioinspired approach designed to solve complex optimization problems [Fuchigami and Abreu, 2023]. Its foundation is rooted in mimicking the expansive growth behavior of fungal mycelium, which is both chaotic and adaptive [Lin et al., 2021]. The algorithm operates through two primary mechanisms: a local chaotic search, inspired by the random spread of fungal hyphae, and a global deterministic search, which mimics the directed nutrient transport towards optimal growth conditions [Alnahwi et al., 2021].

The FKE algorithm, in particular, stands out for its effectiveness in navigating complex spaces by balancing exploration and exploitation [Zhao et al., 2022]. This capability has made it suitable for a wide range of applications, including telecommunications, smart manufacturing, energy optimization, and financial data analysis [Huang et al., 2023]. By simulating natural processes, bioinspired algorithms like FKE draw on millions of years of evolutionary adaptations, allowing them to solve problems more effectively than con-

ventional optimization methods [Lin et al., 2021].

2.1 Contextualization

The Fungi Kingdom Expansion algorithm is part of a broader class of nature-inspired computational techniques [Zhao et al., 2022]. Traditional optimization methods, such as gradient descent or brute force searches, often struggle with high-dimensional problems or those with complex landscapes characterized by multiple local optima [Fuchigami and Abreu, 2023]. Algorithms inspired by biological processes—such as Genetic Algorithms (GAs), Particle Swarm Optimization (PSO), and FKE—have proven to be more flexible and robust, particularly in scenarios where the solution space is vast and difficult to explore comprehensively [Alnahwi et al., 2021].

The adaptive nature of fungi in their natural environment, particularly their ability to expand in response to environmental stimuli and grow toward nutrient-rich areas, is a key feature that is computationally advantageous [Lin et al., 2021]. The chaotic local search component of FKE allows for thorough exploration of the solution space in the early stages of the search, avoiding premature convergence on suboptimal solutions [Fuchigami and Abreu, 2023]. Meanwhile, the deterministic global search phase directs the algorithm’s focus toward the most promising areas, analogous to the movement of resources in fungal networks toward regions that offer the best conditions for growth [Huang et al., 2023].

2.2 Key Methodologies

The FKE algorithm employs a dual search strategy to optimize solutions [Lin et al., 2021]. The first phase, chaotic local search, mirrors the unpredictable spread of fungal hyphae in various directions, allowing for the exploration of different areas of the solution space [Alnahwi et al., 2021]. This helps prevent the algorithm from becoming overly focused on a narrow set of potential solutions too early in the process [Zhao et al., 2022]. The second phase, deterministic global search, is more focused and allows the algorithm to “zoom in” on areas of the solution space that are most promising, based on feedback from the chaotic phase [Huang et al., 2023]. This two-step process is repeated iteratively, improving the chances of finding optimal or near-optimal solutions [Fuchigami and Abreu, 2023].

One of the most significant strengths of the FKE algorithm is its adaptability [Alnahwi et al., 2021]. It can be fine-tuned depending on the problem it is being applied to [Zhao et al., 2022]. For example, the balance between local and global search phases can be adjusted depending on whether the problem requires more exploration or more exploitation [Lin et al., 2021]. Furthermore, FKE incorporates a spore dispersion mechanism, which simulates how fungi reproduce and spread over large distances when local

conditions are not favorable [Fuchigami and Abreu, 2023]. In optimization terms, this mechanism allows the algorithm to explore distant areas of the solution space if the current region does not yield promising results [Huang et al., 2023].

Key Applications

1. **Antenna Radiation Pattern Optimization:** Alnahwi et al. (2021) applied the FKE algorithm to optimize the radiation pattern of antenna arrays [Alnahwi et al., 2021]. Specifically, the chaotic search was used to explore various antenna configurations, while the deterministic phase focused on reducing the side lobes that degrade signal quality [Lin et al., 2021]. The results demonstrated that FKE achieved superior results compared to traditional methods like PSO and GAs, with more effective side lobe reduction and improved overall signal clarity [Fuchigami and Abreu, 2023].
2. **Anti-jamming Systems in Communication Networks:** In 2023, researchers publishing in SciELO explored the use of the FKE algorithm to address interference in communication systems, specifically jamming attacks [Fuchigami and Abreu, 2023]. The algorithm employed chaotic search to quickly identify multiple signal paths, while global search was used to detect and neutralize sources of interference [Alnahwi et al., 2021]. The results showed that this method is faster and more reliable than traditional approaches, making it a valuable tool for both civilian and military communication systems [Lin et al., 2021].
3. **Optimization in Smart Manufacturing:** Zhao et al. (2022) adapted the FKE algorithm for scheduling and resource allocation in smart manufacturing systems [Zhao et al., 2022]. The chaotic phase was employed to explore different production schedules and resource configurations, while the global search focused on minimizing bottlenecks and optimizing resource use [Fuchigami and Abreu, 2023]. The results showed improved production efficiency and reduced operational costs, demonstrating the algorithm’s versatility in industrial applications [Huang et al., 2023].
4. **Energy Consumption Optimization in Data Centers:** Huang et al. (2023) applied the FKE algorithm to minimize energy consumption in data centers [Huang et al., 2023]. The chaotic search was used to explore different configurations for workload distribution and cooling systems, while the deterministic phase optimized energy usage based on real-time demand [Lin et al., 2021]. The results indicated significant reductions in energy consumption, particularly in cooling systems, while maintaining optimal performance of the data center [Fuchigami and Abreu, 2023].

5. Financial Data Analysis and Consumer Behavior: Lin et al. (2021) used FKE to analyze consumer behavior through large-scale credit card transaction datasets [Lin et al., 2021]. The chaotic phase explored broad patterns in consumer spending, while the global search identified specific anomalies or irregular spending behaviors [Alnahwi et al., 2021]. Compared to traditional machine learning approaches, FKE was able to detect trends and anomalies with greater accuracy and speed, making it an invaluable tool for financial institutions [Zhao et al., 2022].

2.3 Results and Advantages

Across these diverse applications, the Fungi Kingdom Expansion algorithm has demonstrated several key advantages over traditional optimization methods [Huang et al., 2023]. Firstly, its ability to balance chaotic exploration with deterministic exploitation makes it particularly well-suited to problems that involve large, complex search spaces [Alnahwi et al., 2021]. Finally, FKE’s flexibility allows it to be adapted to a wide range of applications, from telecommunications and manufacturing to energy management and financial analysis [Lin et al., 2021].

The results from these studies indicate that FKE outperforms traditional optimization algorithms in terms of speed, accuracy, and adaptability [Fuchigami and Abreu, 2023]. In antenna radiation optimization, it reduced interference and improved signal clarity more effectively than Particle Swarm Optimization and Genetic Algorithms [Alnahwi et al., 2021]. In smart manufacturing, it optimized resource allocation and scheduling, resulting in reduced production costs and improved efficiency [Zhao et al., 2022]. Additionally, in the context of energy optimization, FKE minimized power consumption in data centers by dynamically adjusting cooling systems, proving to be more effective than conventional rule-based systems [Huang et al., 2023].

3 Problem Definition

The exponential growth in the use of credit cards has generated large volumes of transactional data containing valuable information about user behavior, from spending habits to consumption preferences. If properly analyzed, this data can help financial institutions make more informed decisions in areas such as fraud detection, credit risk management, and the personalization of offers and services. However, extracting value from these large datasets is not an easy task due to their complexity and the non-linear consumption patterns.

In this context, the objective of this project is to classify and segment banking users based on their use of credit cards. The proposed task is to address this segmentation using clustering techniques, a typical unsupervised learning method, since there are no predefined labels indicating which groups users belong to. Instead of using traditional

clustering methods, this project proposes the development of a bioinspired algorithm based on fungal growth, emulating the mycelial expansion process to identify hidden patterns in the data.

The motivation for using a bioinspired approach lies in the unique adaptive growth characteristics that fungi exhibit in nature. The primary goal is to maximize the quality of classification through a process that follows biological principles, leveraging the decentralized and self-organized nature of mycelia. This approach could offer an advantage over traditional clustering algorithms by capturing the complex and stochastic patterns present in user data. However, the random and adaptive behavior of mycelial expansion presents challenges in terms of stability and consistency of the clusters, which could affect the results obtained.

The dataset used for this project includes approximately 8,950 user records, each described by 17 features detailing various aspects of credit card usage, such as account balance, purchase amounts, and credit limits, among other important characteristics. This dataset is representative of customers’ financial behavior, and without a target variable, the task is to automatically uncover latent structures that enable the formation of coherent groups.

To evaluate the quality of the clustering, three standard metrics will be used: the Davies-Bouldin index, the silhouette score, and the Dunn index. The Davies-Bouldin index will assess the relationship between cluster compactness and separation, providing an effective measure of the internal cohesion of the groups. The silhouette score will measure how well points within a cluster are grouped compared to others, offering insight into the proximity of points to their own clusters versus others. Finally, the Dunn index will help measure the distance between the closest clusters and the dispersion of the widest cluster, providing a reference for the separation between groups and their distribution in feature space.

An inherent limitation of the bioinspired approach is its stochastic behavior, which simulates the adaptive and decentralized properties of mycelial growth. This characteristic introduces a degree of variability in the results, which could affect the exact reproducibility of the clusters in each run of the algorithm. While this stochasticity can be a strength in terms of flexibility and adaptability to the data, it also presents a challenge when trying to ensure consistent results. The impact of this feature will be considered throughout the experiments and evaluations using the selected metrics.

In summary, this project aims to combine clustering techniques with bioinspired principles of fungal growth to identify usage patterns among banking users, enabling them to be grouped into clearly differentiated segments.

4 Dataset Presentation

In this project we will use the following dataset: **Credit Card Dataset for Clustering** from Kaggle [Bhasin, 2013]. This dataset contains information on more than 8,000 credit card users with 18 characteristics including total balance, frequency of purchases, credit limit, number of months the customer has had the card, among others.

It was found that there is a column with more than 300 missing values, which is `minimum_payment`, the minimum payment to be made by the customer. However, since the percentage of missing values in this column is very low relative to the total size of the dataset, we have decided to ignore it to simplify the analysis and avoid additional complications.

It is observed that there are variables with a strong correlation between each other, such as between the characteristics `purchases` and `oneoff_purchases`, see Figure 1:

Table 1: Statistical Summary of Variables

Variable	Count	Mean	Min	Max
BALANCE FREQUENCY	8950	0.88	0.00	1.00
PURCHASES	8950	1003.20	0.00	49039.57
ONEOFF PURCHASES	8950	592.44	0.00	40761.25
INSTALLMENTS PURCHASES	8950	411.07	0.00	22500.00
CASH ADVANCE PURCHASES	8950	978.87	0.00	47137.21
PURCHASES FREQUENCY	8950	0.49	0.00	1.00
ONEOFF PURCHASES FREQ	8950	0.20	0.00	1.00
PURCHASES INSTALL FREQ	8950	0.36	0.00	1.00
CASH ADVANCE FREQ	8950	0.14	0.00	1.50
CASH ADVANCE TRX	8950	3.25	0.00	123.00
PURCHASES TRX	8950	14.71	0.00	358.00
CREDIT LIMIT	8949	4494.45	50.00	30000.00
PAYMENTS	8637	1733.14	0.00	50721.48
MINIMUM PAYMENTS	8637	864.21	0.02	76406.21
PRC FULL PAYMENT	8950	0.15	0.00	1.00
TENURE	8950	11.52	6.00	12.00

5 Strategy for Solving the Problem

The strategy for developing and implementing the FKE focuses on several key components that will be essential to ensure that FKE provides a robust and effective solution for clustering credit card users. This strategy aims not only to improve the accuracy in identifying consumption patterns but also to provide an innovative tool capable of adapting to the complexities and changing dynamics of financial data.

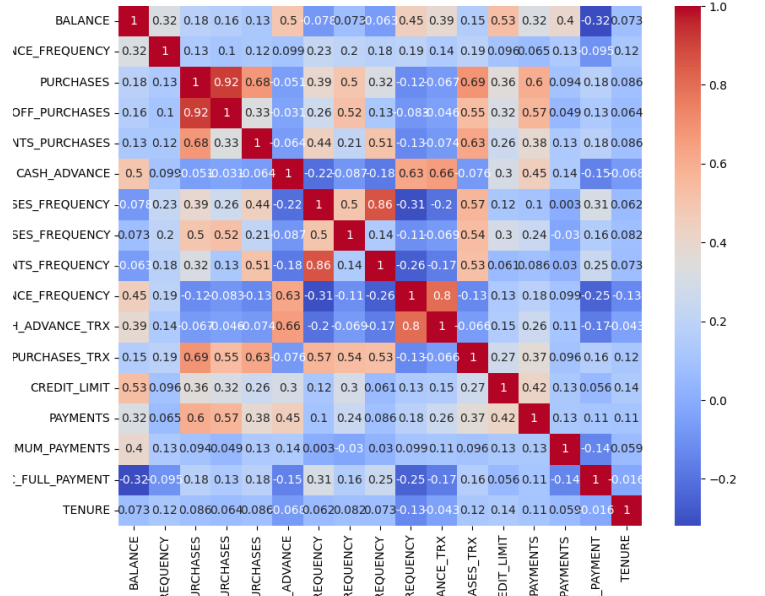


Figure 1: Correlation Matrix of characteristics on the dataset

The different stages of the strategy to be implemented are as follows:

1. **Data Analysis and Preparation:** The first fundamental step is data collection and preparation. Raw data must undergo preprocessing, including cleaning and normalization. This process is essential to ensure the quality and consistency of the data, which involves handling missing values, removing outliers, and normalizing variables to facilitate analysis. Next, a data exploration phase is conducted through exploratory analysis to better understand the characteristics of the data and detect preliminary patterns that can guide the development of the algorithm.
2. **Development of the Fungi Kingdom Expansion (FKE) Algorithm:** With the data prepared, the next step is the development of the Fungi Kingdom Expansion (FKE) algorithm. Inspired by the chaotic and deterministic expansion of fungal mycelium, the algorithm's design must reflect two main components: first, the stationary mass representing the structure of the hyphae that extends deterministically in search of resources; and second, the mobile mass that simulates the flow of nutrients to the optimal tips of the hyphae, promoting global exploration and adaptation. The algorithm must incorporate search mechanisms that combine chaotic (local) and deterministic (global) strategies to optimize user clustering. The implementation must consider how FKE can effectively balance

local and global exploration to identify significant patterns in the data.

3. **Algorithm Configuration and Execution:** Once the algorithm is developed, the next step is configuration and execution. This includes defining and adjusting the key parameters of FKE, such as the expansion rate, the balance between local and global exploration, and convergence criteria. The training phase involves applying the algorithm to a subset of data to fine-tune and optimize its performance. During this phase, iterative testing is necessary to improve the algorithm's effectiveness. Cross-validation is crucial for assessing the robustness of the model, ensuring that FKE can handle different datasets and maintain clustering accuracy.
4. **Evaluation of Results:** Evaluating the results is a key component for measuring the algorithm's success. Evaluation metrics such as the Silhouette index, Davies-Bouldin coefficient, and cluster homogeneity should be used to assess clustering quality. This analysis will help determine the effectiveness of FKE and qualitatively evaluate the model and its feasibility in this context.

nutrient sources. This balance between exploration (chaotic) and exploitation (directed) is fundamental to the algorithm's effectiveness.

3. **Convergence:** The algorithm checks for convergence of the fungal colonies by comparing the previous and current centroid positions. If the change in positions is less than a defined tolerance, the algorithm considers that convergence has been reached and stops.

6 Definition of the Model

The Fungi Kingdom Expansion (FKE) algorithm is based on the natural behavior of fungal colonies and their ability to adapt and grow in search of resources. This algorithm operates on two main phases: local expansion and global adjustment, which simulate the behavior of fungi as they explore and colonize their environment. Below is a detailed description of the model's characteristics and functionality.

1. **Local Expansion (Chaotic):** In the first phase, the model simulates the chaotic expansion of fungal spores. Each point in the dataset is considered a "nutrient patch," and the algorithm assigns these patches to the nearest fungal colonies (centroids). During this process, random noise is introduced into the distance calculations, simulating the unpredictable nature of fungal growth. This perturbation helps the model escape local optima and explore new areas within the data space, reflecting how fungi might disperse their spores in search of favorable conditions.
2. **Global Adjustment (Directed):** After the local expansion phase, the model enters the global adjustment phase. In this stage, the positions of the fungal colonies are recalculated based on the nutrient patches they are assigned to. The new centroid of each colony is determined by calculating the average positions of the assigned points, with a global adjustment that simulates how fungi relocate themselves toward rich

Algorithm 1 FungiKingdomExpansion Algorithm

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1: Input:  $n\_colonies$ ,  $max\_growth\_cycles$ ,  $local\_growth\_range$ ,
    $global\_growth\_range$ ,  $growth\_tolerance$ 
2: Output: Colony centers (centroids) and spore cluster labels
   FungiInspiredClustering $n\_colonies$ ,  $max\_growth\_cycles$ ,
    $local\_growth\_range$ ,  $global\_growth\_range$ ,  $growth\_tolerance$ 
3: function GROW( $self$ ,  $NutrientField$ )
4:   Initialize spore colonies by randomly placing  $n\_colonies$  on the nu-
   trient field ( $NutrientField$ )
5:   for  $cycle \leftarrow 1$  to  $max\_growth\_cycles$  do
6:     Phase 1: Local Expansion (Chaotic Growth)
7:     Assign each nutrient point to the closest colony with local dis-
     turbance (noise)
8:      $spore\_distribution \leftarrow self.ASSIGN\_LOCAL\_COLONIES(NutrientField)$ 
9:     Phase 2: Global Colony Repositioning (Directed
Growth)
10:    Store current colony centers as  $previous\_colonies$ 
11:    Update colony positions to reflect nutrient-rich areas
12:     $colony\_centers \leftarrow self.REPOSITION\_COLONIES(NutrientField)$ 
13:    Convergence Check
14:    if  $self.GROWTH\_STABILIZED(previous\_colonies, colony\_centers)$ 
then
15:      break ▷ Stop growth if colony centers stabilize
16:    end if
17:  end for
18: end function
19: function ASSIGN_LOCAL_COLONIES( $self$ ,  $NutrientField$ )
20:   for each  $nutrient\_patch$  in  $NutrientField$  do
21:     Compute distance between  $nutrient\_patch$  and each colony cen-
     ter
22:     Add local noise to the distances (simulating random fungal
     growth)
23:     Assign  $nutrient\_patch$  to the nearest colony based on noisy dis-
     tances
24:   end for
25:   return  $spore\_distribution$ 
26: end function
27: function REPOSITION_COLONIES( $self$ ,  $NutrientField$ )
28:   for each colony  $c$  do
29:     Gather nutrient patches assigned to colony  $c$ 
30:     if colony  $c$  has nutrient patches then
31:       Compute new colony center by averaging nutrient patches
       with global adjustment
32:     else
33:       Relocate the colony center to a random nutrient-rich patch
34:     end if
35:     Add the updated colony center to the list
36:   end for
37:   return  $colony\_centers$ 
38: end function
39: function GROWTH_STABILIZED( $self$ ,  $previous\_colonies$ ,  $colony\_centers$ )
40:   Compute distance between old and new colony centers
41:   return True if the maximum distance is less than
    $growth\_tolerance$ 
42: end function

```

7 References

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